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Graded Quiz • 30 min

English Due Jun 9, 11:59 PM IST

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Nonparametric Regression: Theory

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Nonparametric Regression: Data Analysis

Assignments: Nonparametric Regression and Smoothing Functions

Nonparametric Regression: Theory

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Review Learning Objectives

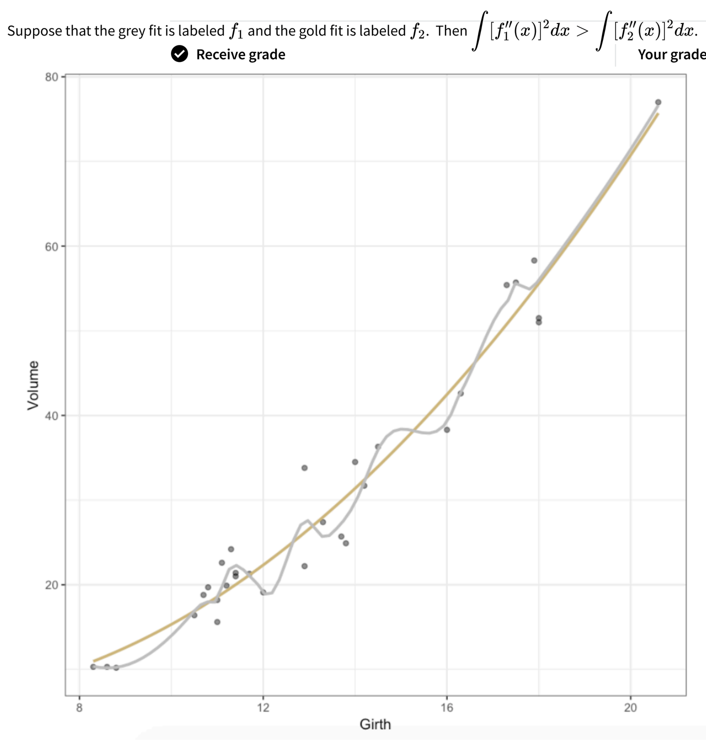
1. The trees data frame has 31 observations on 3 variables. **3 / 3 points**

1. **Girth**: numeric Tree diameter in inches

2. **Height**: numeric Height in ft

3. **Volume**: numeric Volume of timber in cubic ft

Suppose that the grey fit is labeled f_1 and the gold fit is labeled f_2. Then $\int [f_1''(x)]^2 dx > \int [f_2''(x)]^2 dx$.



- (•) True
- ( ) False

Correct

2. Parametric modeling is more efficient when the relationship between variables is unknown. **3 / 3 points**

- ( ) True
- (•) False

Correct

3. Binomial regression is a type of nonparametric model. **3 / 3 points**

- ( ) True
- (•) False

Correct

4. In the context of kernel estimation, the smaller the bandwidth, the rougher the fit. **3 / 3 points**

- (•) True
- ( ) False

Correct

5. Consider the following modeling scenario: Ethanol fuel was burned in a single-cylinder engine. For various settings of the engine compression, the emissions of nitrogen oxides were recorded: **4 / 4 points**

NOx: Concentration of nitrogen oxides (NO and NO2) in micrograms/J).

C: Compression ratio of the engine.

Researchers would like to understand how NOx is related to C. It is quite plausible that the relationship is nonlinear.

Based on the information given, a reasonable first attempt at answering this question would include:

- [ ] A linear regression model.
- [x] A smoothing spline.

  Correct
- [x] A kernel regreesion.

  Correct
- [x] A loess model.

  Correct
- [ ] A Poisson regression model.

6. The nonparametric approach assumes far less about the form of the model and so it is less liable to make major mistakes that result in bias. **3 / 3 points**

- (•) True
- ( ) False

Correct

7. Nonparametric models often don't have a formulaic way of describing the relationship between the predictors and response. **3 / 3 points**

- (•) True
- ( ) False

Correct

8. In smoothing spline regression, we estimate our model f by minimizing the mean squared error: **3 / 3 points**

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - f(x_i))^2.$$

- ( ) True
- (•) False

Correct

9. In the context of smoothing spline regression, as $\lambda \to \infty$, the fit converges to $\hat{f}(x_i) = 0$ for all i. **3 / 3 points**

- (•) True
- ( ) False

Correct

10. Since loess models rely on theory for weighted regression, it is not possible to quantify uncertainties in the model, in much the same way as is done for, e.g., linear regression. **3 / 3 points**

- ( ) True
- (•) False

Correct

11. Disadvantages of the loess fit include: **4 / 4 points**

- [ ] Loess requires a pre-specified functional form.
- [x] Loess is not as easy to interpret as standard linear regression.

  Correct
- [x] Loess can be computationally expensive.

  Correct
- [x] Loess requires fairly large, densely sampled data in order to produce good models.

  Correct